Jerrold E. Marsden and Anthony J. Tromba

Vector Calculus

Fifth Edition

Chapter 3:
High-Order Derivatives: Maxima and Minima

3.3 Extrema of Real-Valued Functions

3.3 Extrema of Real Valued Functions

Key Points in this Section.

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1. **Definitions.** A ***local minimum point*** of $f : U \subset \mathbb{R}^n \to \mathbb{R}$ is a point $\mathbf{x}_0 \in U$ such that $f(\mathbf{x}_0) \leq f(\mathbf{x})$ for all $\mathbf{x}$ in some neighborhood of $\mathbf{x}_0$; we say $f(\mathbf{x}_0)$ is the corresponding ***local minimum value***. If, similarly, $f(\mathbf{x}_0) \geq f(\mathbf{x})$, then $\mathbf{x}_0$ is a ***local maximum point*** (and $f(\mathbf{x}_0)$ is the ***local maximum value***). If $\mathbf{x}_0$ is either of these, it is a ***local extremum***.

2. **First Derivative Test.** If $U \subset \mathbb{R}^n$ is open, $f : U \subset \mathbb{R}^n \to \mathbb{R}$ is differentiable and $\mathbf{x}_0$ is a local extremum, then $\mathbf{x}_0$ is a ***critical point***; that is, all the partials of f vanish at $\mathbf{x}_0$:

$$\frac{\partial f}{\partial x_1}(\mathbf{x}_0) = 0, \cdots, \frac{\partial f}{\partial x_0}(\mathbf{x}_0) = 0.$$

The idea of the proof is to apply the one-variable first derivative test to f restricted to lines through $\mathbf{x}_0$.

3. If $f : U \subset \mathbb{R}^n \to \mathbb{R}$ is C^2, the ***Hessian*** of f at $\mathbf{x}_0$ is the quadratic function of $\mathbf{h}$ given by

$$Hf(\mathbf{x}_0)(\mathbf{h}) = \frac{1}{2}[h_1, \ldots, h_n] \begin{bmatrix} \dfrac{\partial^2 f}{\partial x_1 \partial x_1} & \cdots & \dfrac{\partial^2 f}{\partial x_1 \partial x_n} \\ \vdots & & \vdots \\ \dfrac{\partial^2 f}{\partial x_n \partial x_1} & \cdots & \dfrac{\partial^2 f}{\partial x_n \partial x_n} \end{bmatrix} \begin{bmatrix} h_1 \\ \vdots \\ h_n \end{bmatrix}$$

which also equals the second term in the Taylor expansion of f about $\mathbf{x}_0$.

4. **Second Derivative Test—n Variables.** If $f : U \subset \mathbb{R}^n \to \mathbb{R}$ is C^3 (and again U is open), $\mathbf{x}_0$ is a critical point, and if $Hf(\mathbf{x}_0)(\mathbf{h}) > 0$ for all $\mathbf{h} \neq \mathbf{0}$ (that is, $Hf(\mathbf{x}_0)$ is ***positive definite***), then $\mathbf{x}_0$ is a local minimum. Likewise, if $Hf(\mathbf{x}_0)(\mathbf{h}) < 0$ for all $\mathbf{h} \neq \mathbf{0}$, (that is, $Hf(\mathbf{x}_0)$ is ***negative definite***), then $\mathbf{x}_0$ is a local maximum.

5. The idea of the proof of the second derivative test is to apply the second order Taylor theorem and show that the remainder term can be ignored.

6. **Second Derivative Test—Two Variables.** Let $f : U \subset \mathbb{R}^2 \to \mathbb{R}$ (again with U open) be of class C^3. A point $(x_0, y_0) \in U$ is a local minimum if the following conditions are satisfied:

(i) $\dfrac{\partial f}{\partial x}(x_0, y_0) = \dfrac{\partial f}{\partial y}(x_0, y_0) = 0$ (that is, (x_0, y_0) is a critical point)

(ii) $\dfrac{\partial^2 f}{\partial x^2}(x_0, y_0) > 0$

(iii) $D = \begin{vmatrix} \dfrac{\partial^2 f}{\partial x^2}(x_0, y_0) & \dfrac{\partial^2 f}{\partial x \partial y}(x_0, y_0) \\ \dfrac{\partial^2 f}{\partial x \partial y}(x_0, y_0) & \dfrac{\partial^2 f}{\partial y^2}(x_0, y_0) \end{vmatrix} > 0.$

If (i) and (iii) hold, but $\partial^2 f / \partial x^2$ at (x_0, y_0) is negative, then (x_0, y_0) is a local maximum. If the ***discriminant*** D is negative, then (x_0, y_0) is a ***saddle point*** (that is, (x_0, y_0) is neither a local maximum nor a local minimum).

7. **Global Extrema.** Let $f : A \subset \mathbb{R}^n \to \mathbb{R}$, where A need not be open. A point $\mathbf{x}_0 \in A$ is an ***absolute*** or ***global minimum*** of f if $f(\mathbf{x}_0) \leq f(\mathbf{x})$ for all $\mathbf{x} \in A$. Similarly, $\mathbf{x}_0$ is an ***absolute*** or ***global maximum*** if $f(\mathbf{x}_0) \geq f(\mathbf{x})$ for all $\mathbf{x} \in A$.

8. If $D \subset \mathbb{R}^n$ is ***closed*** (that is, all boundary points of D lie in D) and ***bounded*** (that is, D is a subset of some, perhaps large ball), and if $f : D \subset \mathbb{R}^n \to \mathbb{R}$ is continuous, then f has (at least one) absolute maximum point $\mathbf{x}_0 \in D$ and (at least one) absolute minimum point $\mathbf{x}_1 \in D$.

9. **Strategy for Global Extrema.** To find absolute extrema on a closed and bounded region $D \subset \mathbb{R}^n$ that is an open set U together with its boundary points ∂U,

(i) find the critical points in U

(ii) find the maximum points of f on ∂U

(iii) compute the values of f at all the points in (i) and (ii)

(iv) the largest such value gives the maximum and the smallest the minimum.

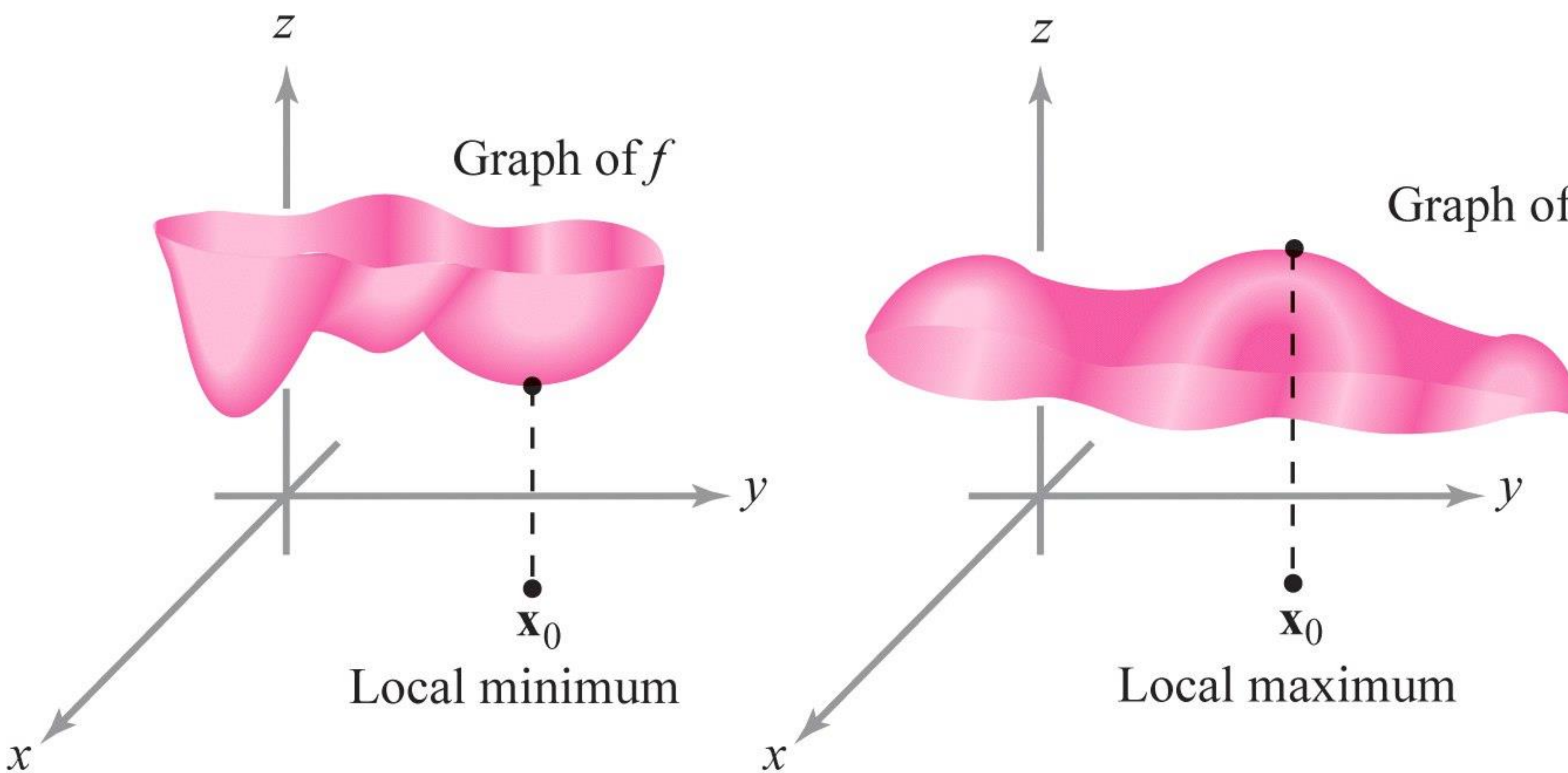
If $n = 2$ and ∂U is a closed curve, step (ii) can be done by parametrizing this curve and using the methods of one-variable calculus. Alternatively, for $n = 2$ or 3, one can use the Lagrange multipliers given in the next section.

Pierre-Louis de Maupertuis
(1698--1759)

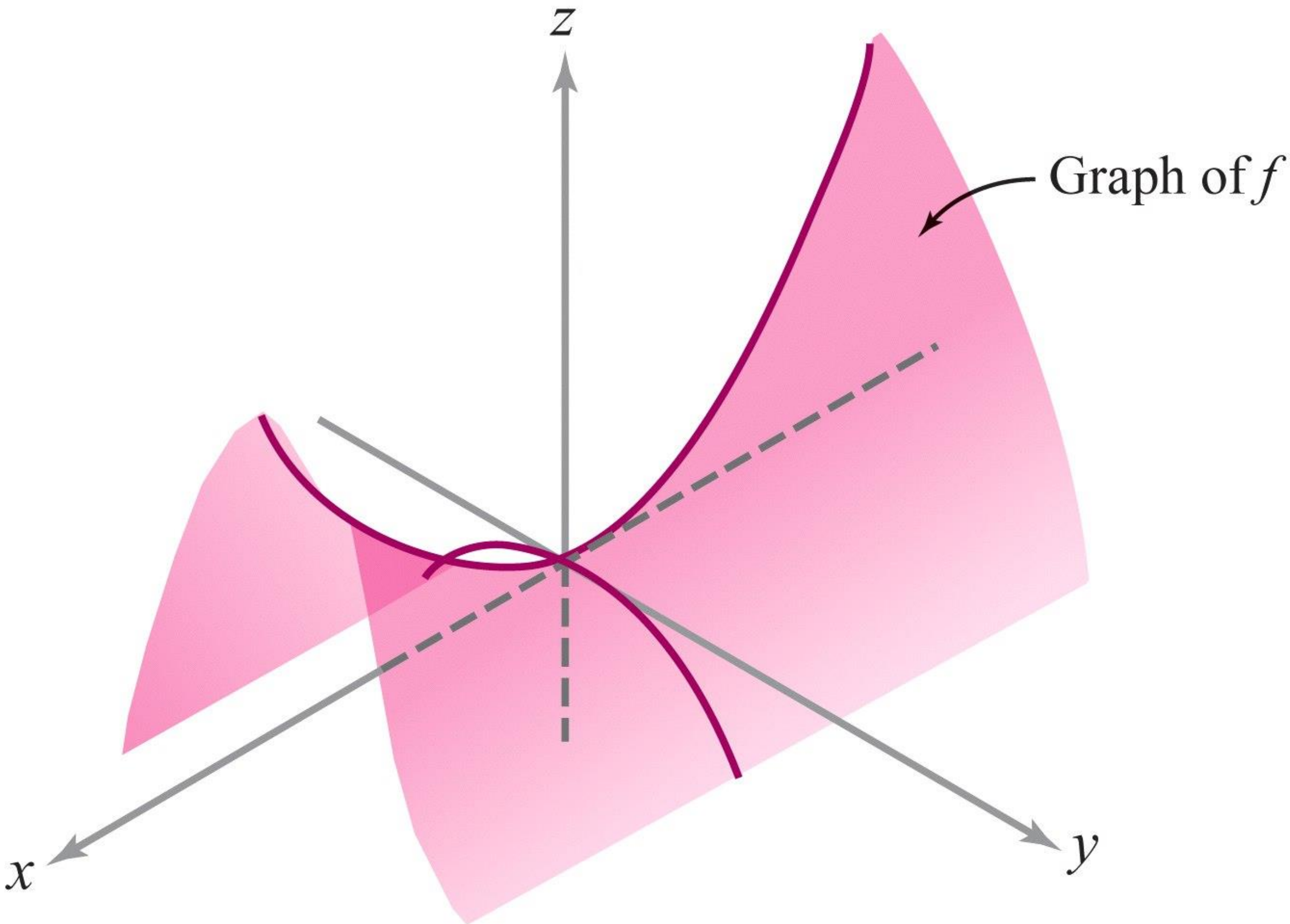
Maupertuis principle
Nature always minimizes action

What satisfaction for the human spirit that, in contemplating these laws which contain the principle of motion and of rest for all bodies in the universe, he finds the proof of existence of Him who governs the world.

Definition If $f\colon U \subset \mathbb{R}^n \to \mathbb{R}$ is a given scalar function, a point $\mathbf{x}_0 \in U$ is called a ***local minimum*** of f if there is a neighborhood V of $\mathbf{x}_0$ such that for all points $\mathbf{x}$ in V, $f(\mathbf{x}) \geq f(\mathbf{x}_0)$. (See Figure 3.3.2.) Similarly, $\mathbf{x}_0 \in U$ is a ***local maximum*** if there is a neighborhood V of $\mathbf{x}_0$ such that $f(\mathbf{x}) \leq f(\mathbf{x}_0)$ for all $\mathbf{x} \in V$. The point $\mathbf{x}_0 \in U$ is said to be a ***local***, or ***relative***, ***extremum*** if it is either a local minimum or a local maximum. A point $\mathbf{x}_0$ is a ***critical point*** of f if either f is *not differentiable at* $\mathbf{x}_0$, or if it is, $\mathbf{D}f(\mathbf{x}_0) = \mathbf{0}$. A critical point that is not a local extremum is called a ***saddle point***.[4]


z
Graph of f
y
$\mathbf{x}_0$
Local minimum
x
z
Graph of f
y
$\mathbf{x}_0$
Local maximum
x

(a) (b)



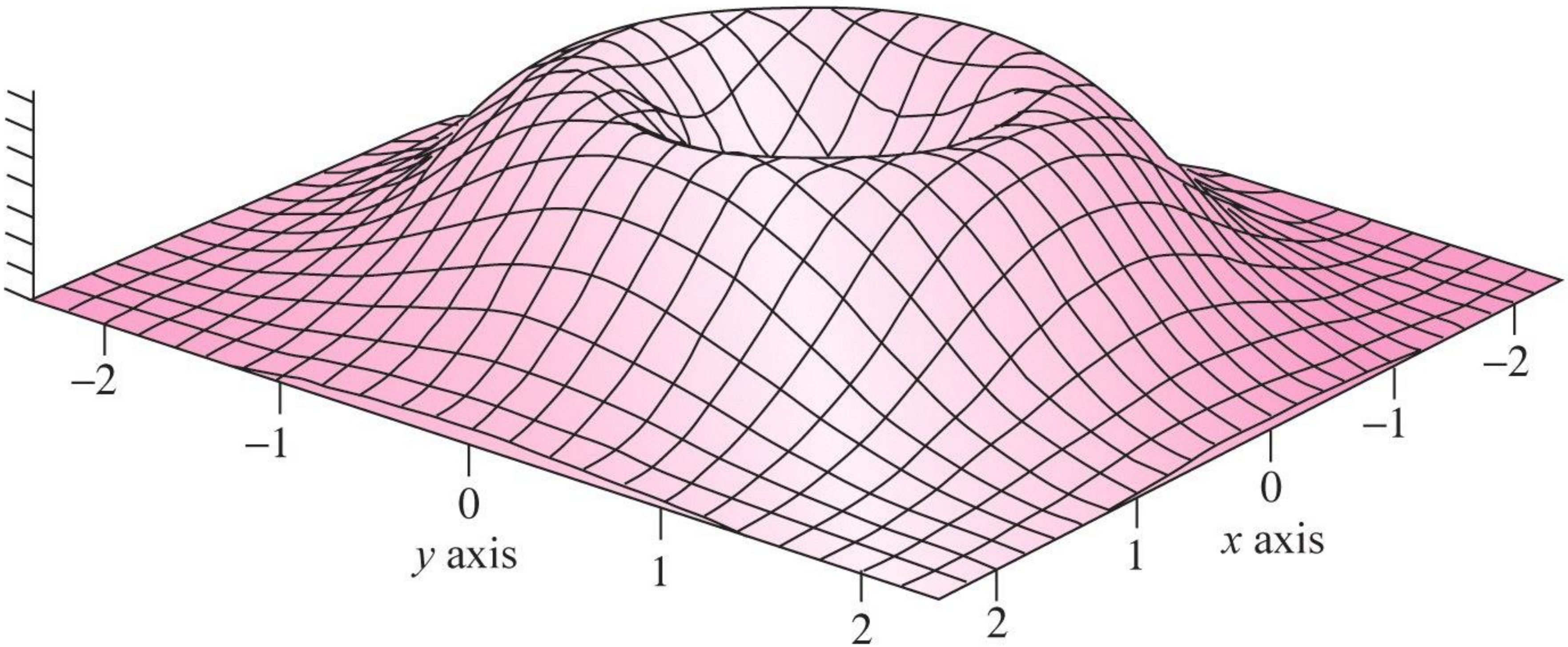
Saddle point

THEOREM 4: First Derivative Test for Local Extrema If $U \subset \mathbb{R}^n$ is open, the function $f: U \subset \mathbb{R}^n \to \mathbb{R}$ is differentiable, and $\mathbf{x}_0 \in U$ is a local extremum, then $\mathbf{D}f(\mathbf{x}_0) = \mathbf{0}$; that is, $\mathbf{x}_0$ is a critical point of f.

Trobeu els punts crítics de:

a) $f(x,y) = x^2 + y^2$
b) $f(x,y) = x^2 - y^2$
c) $f(x,y) = x^2y + y^2x$
d) $f(x,y) = 2(x^2 + y^2)e^{-(x^2+y^2)}$

$$z = 2(x^2 + y^2)\ e^{-x^2-y^2}$$



Una **forma quadràtica** és una funció $g: \mathbb{R}^n \longrightarrow \mathbb{R}$ de la forma

$$g(x_1, \dots, x_n) = \sum_{i=1}^{n} \sum_{j=1}^{n} a_{ij} x_i x_j$$

amb $a_{ij} \in \mathbb{R}$

Les formes quadràtiques són funcions *homogènies de grau 2*

$$g(\lambda x_1, \dots, \lambda x_n) = \lambda^2 g(x_1, \dots, x_n)$$

DEFINITION Suppose that $f\colon U \subset \mathbb{R}^n \to \mathbb{R}$ has second-order continuous derivatives $(\partial^2 f/\partial x_i\,\partial x_j)(\mathbf{x}_0)$, for $i, j = 1, \ldots, n$, at a point $\mathbf{x}_0 \in U$. The ***Hessian of f at $\mathbf{x}_0$*** is the quadratic function defined by

$$
\begin{aligned}
Hf(\mathbf{x}_0)(\mathbf{h}) &= \frac{1}{2}\sum_{i,j=1}^{n} \frac{\partial^2 f}{\partial x_i\,\partial x_j}(\mathbf{x}_0) h_i h_j \\
&= \frac{1}{2}(h_1, \ldots, h_n)\begin{pmatrix} \dfrac{\partial^2 f}{\partial x_1\,\partial x_1} & \cdots & \dfrac{\partial^2 f}{\partial x_1\,\partial x_n} \\ \vdots & & \\ \dfrac{\partial^2 f}{\partial x_n\,\partial x_1} & \cdots & \dfrac{\partial^2 f}{\partial x_n\,\partial x_n} \end{pmatrix}\begin{pmatrix} h_1 \\ \vdots \\ h_n \end{pmatrix}.
\end{aligned}
$$

The matrix H of second derivatives is known as the ***Hessian matrix***.
The Hessian matrix is symmetric.

Sigui una funció $g\colon \mathbb{R}^n \longrightarrow \mathbb{R}$

- g és ***definida positiva*** si $g(\boldsymbol{x}) > 0, \ \ \forall \boldsymbol{x} \in \mathbb{R}^n$
- g és ***definida negativa*** si $g(\boldsymbol{x}) < 0, \ \ \forall \boldsymbol{x} \in \mathbb{R}^n$
- g és ***semi-definida positiva*** si $g(\boldsymbol{x}) \geq 0, \ \ \forall \boldsymbol{x} \in \mathbb{R}^n$
- g és ***semi-definida negativa*** si $g(\boldsymbol{x}) \leq 0, \ \ \forall \boldsymbol{x} \in \mathbb{R}^n$

THEOREM 5: Second Derivative Test for Local Extrema If $f : U \subset \mathbb{R}^n \to \mathbb{R}$ is of class C^3, $\mathbf{x}_0 \in U$ is a critical point of f, and the Hessian $Hf(\mathbf{x}_0)$ is positive-definite, then $\mathbf{x}_0$ is a relative minimum of f. Similarly, if $Hf(\mathbf{x}_0)$ is negative-definite, then $\mathbf{x}_0$ is a relative maximum.

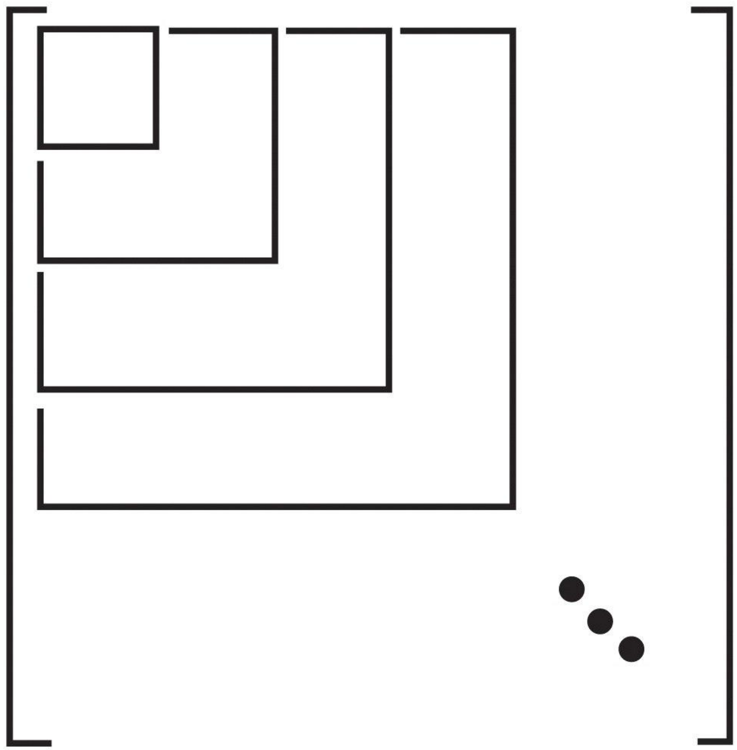
THEOREM 6: Second Derivative Maximum-Minimum Test for Functions of Two Variables Let $f(x, y)$ be of class C^3 on an open set U in $\mathbb{R}^2$. A point (x_0, y_0) is a (strict) local minimum of f provided the following three conditions hold:

(i) $\dfrac{\partial f}{\partial x}(x_0, y_0) = \dfrac{\partial f}{\partial y}(x_0, y_0) = 0$

(ii) $\dfrac{\partial^2 f}{\partial x^2}(x_0, y_0) > 0$

(iii) $D = \left(\dfrac{\partial^2 f}{\partial x^2}\right)\left(\dfrac{\partial^2 f}{\partial y^2}\right) - \left(\dfrac{\partial^2 f}{\partial x\, \partial y}\right)^2 > 0$ at (x_0, y_0)

(D is called the ***discriminant*** of the Hessian.) If in (ii) we have < 0 instead of > 0 and condition (iii) is unchanged, then we have a (strict) local maximum.



Sigui A una matriu simètrica

Siguin $D_1, D_2, \dots, D_n$ els determinants d'aquestes submatrius principals de A

Criteri de Sylvester

A és ***definida positiva*** sii $D_i > 0, \forall i$

A és ***definida negative*** sii $D_1, D_3, \dots < 0$ i $D_2, D_4, \dots > 0$

second partials Test

Let $z = f(x, y)$ be a function of two variables for which the first- and second-order partial derivatives are continuous on some disk containing the point (x_0, y_0). Suppose $f_x(x_0, y_0) = 0$ and $f_y(x_0, y_0) = 0$. Define the quantity

$$D = f_{xx}(x_0, y_0) f_{yy}(x_0, y_0) - \left(f_{xy}(x_0, y_0)\right)^2. \tag{13.8.8}$$

Then:

i. If $D > 0$ and $f_{xx}(x_0, y_0) > 0$, then f is concave up at this critical point, so f has a local minimum at (x_0, y_0).
ii. If $D > 0$ and $f_{xx}(x_0, y_0) < 0$, then f is concave down at this critical point, so f has a local maximum at (x_0, y_0).
iii. If $D < 0$, then f has a saddle point at (x_0, y_0).
iv. If $D = 0$, then the test is inconclusive.

See Figure 13.8.4.

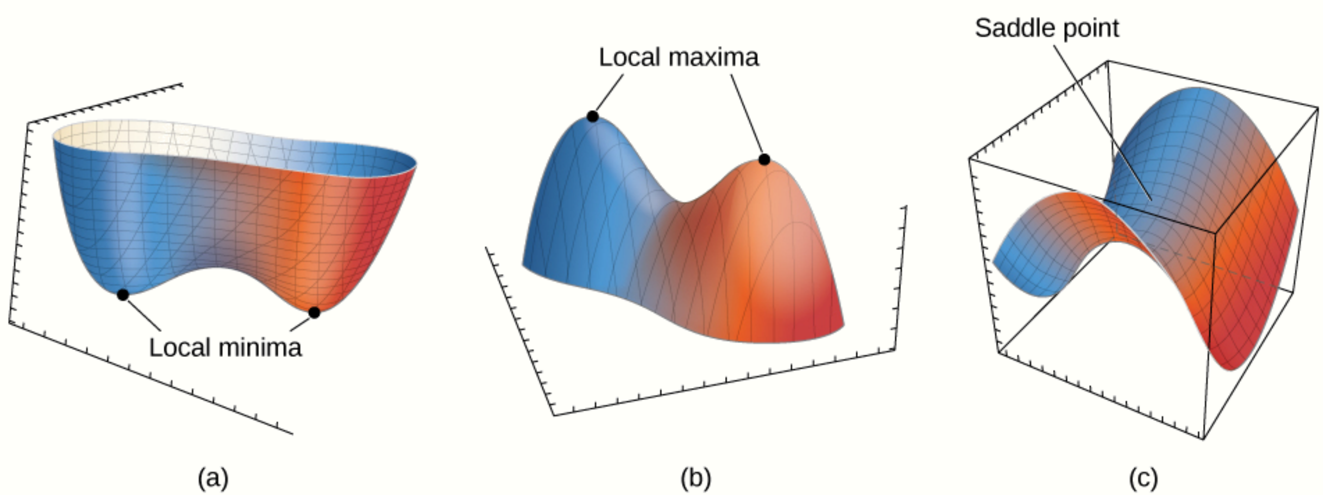
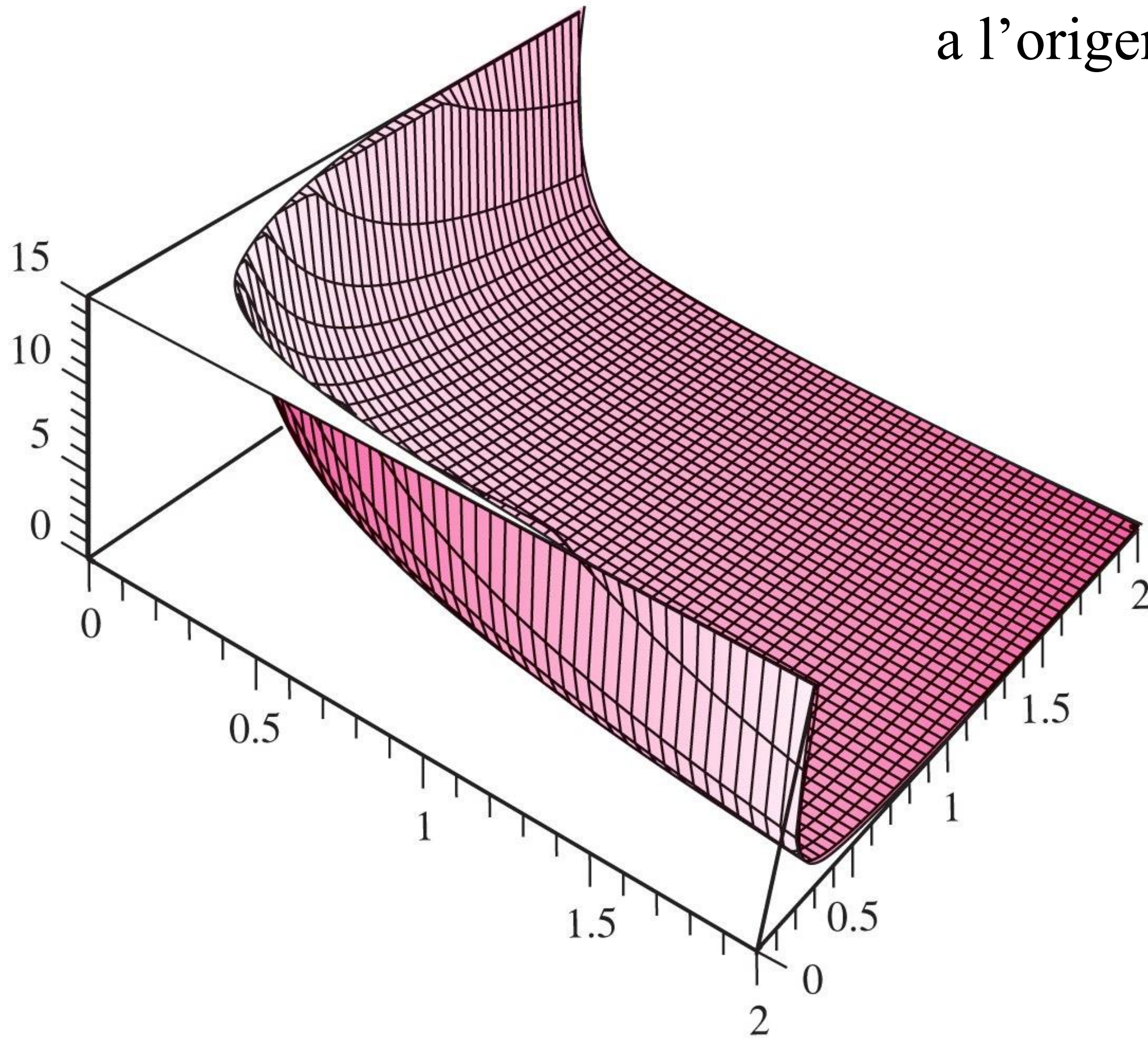


Figure 13.8.4: The second partials test can often determine whether a function of two variables has a local minima (a), a local maxima (b), or a saddle point (c).

Analitzeu els punts crítics de:

a) $f(x,y) = x^2 - 2xy + 2y^2$
b) $f(x,y) = \ln(1 + x^2 + y^2)$
c) $f(x,y) = x^5y + xy^5 + xy$

Analitzeu la funció $f(x,y,z) = x^2 + y^2 + z^2 + 2xyz$ en els punts $(0,0,0)$ i $(-1,1,1)$



Trobar els punts més propers a l'origen de la superfície

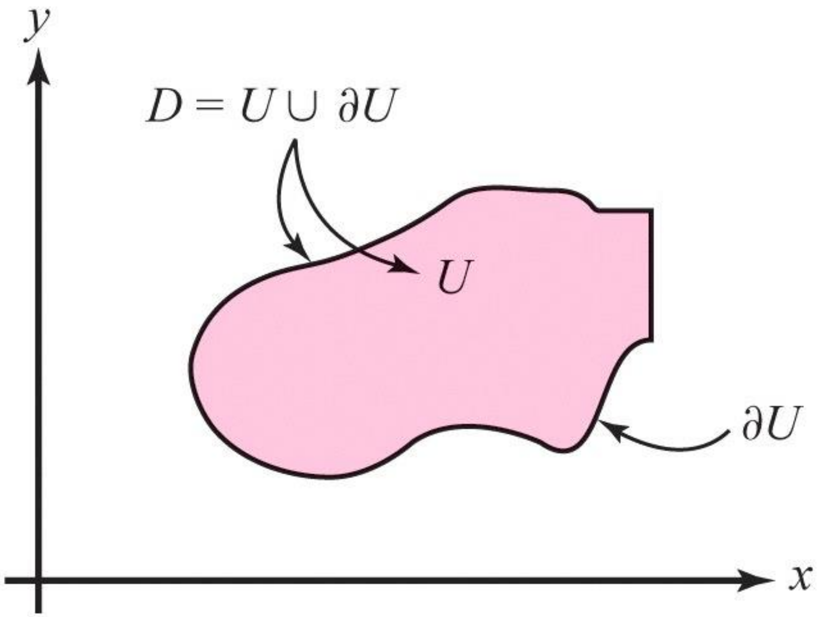
$$z = \frac{1}{xy}$$

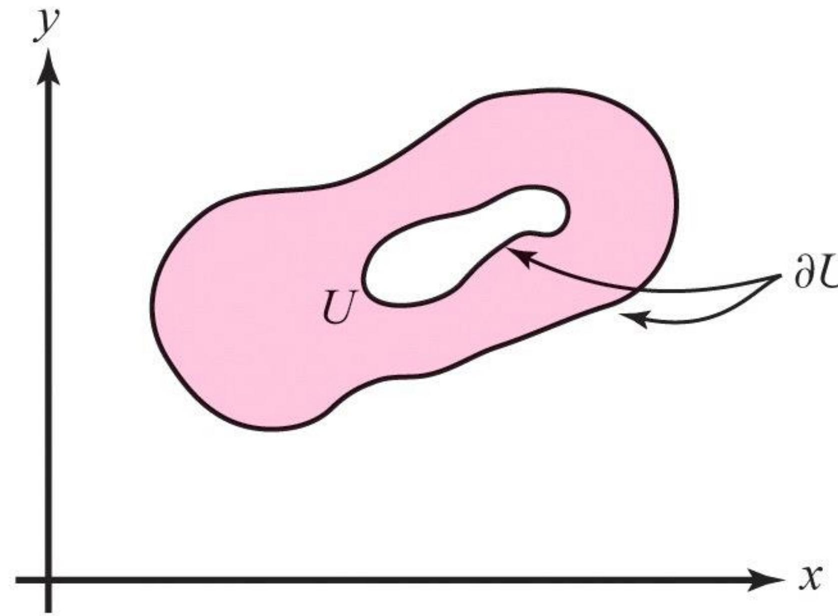
DEFINITION Suppose $f\colon A \to \mathbb{R}$ is a function defined on a set A in $\mathbb{R}^2$ or $\mathbb{R}^3$. A point $\mathbf{x}_0 \in A$ is said to be an ***absolute maximum*** (or ***absolute minimum***) point of f if $f(\mathbf{x}) \leq f(\mathbf{x}_0)$ [or $f(\mathbf{x}) \geq f(\mathbf{x}_0)$] for all $\mathbf{x} \in A$.

DEFINITION A set $D \in \mathbb{R}^n$ is said to be ***bounded*** if there is a number $M > 0$ such that $\|\mathbf{x}\| < M$ for all $\mathbf{x} \in D$. A set is ***closed*** if it contains all its boundary points.

THEOREM 7: Global Existence Theorem for Maxima and Minima

Let D be closed and bounded in $\mathbb{R}^n$ and let $f\colon D \to \mathbb{R}$ be continuous. Then f assumes its absolute maximum and minimum values at some points $\mathbf{x}_0$ and $\mathbf{x}_1$ of D.


y
$D = U \cup \partial U$
U
∂U
x


y
U
∂U
x

Strategy for Finding the Absolute Maxima and Minima on a Region with Boundary Let f be a continuous function of two variables defined on a closed and bounded region D in $\mathbb{R}^2$, which is bounded by a smooth closed curve. To find the absolute maximum and minimum of f on D:

(i) Locate all critical points for f in U.

(ii) Find the maximum and minimum of f viewed as a function only on ∂U.

(iii) Compute the value of f at all of these critical points.

(iv) Compare all these values and select the largest and the smallest.

Troba els màxims i mínims absoluts de

$$f(x, y) = x^2 + y^2 - x - y + 1$$

en el disc D definit per $x^2 + y^2 \leq 1$

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3.4 Constrained Extrema and Lagrange Multipliers

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Key Points in this Section.

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1. **Lagrange Multiplier Equations.** Let $f : U \subset \mathbb{R}^n \to \mathbb{R}$ and $g : U \subset \mathbb{R}^n \to \mathbb{R}$ be C^1. Consider the problem of extremizing f on a level set of g, say $g(\mathbf{x}) = c$. If $\mathbf{x}_0$ is such an extremum and if $\nabla g(\mathbf{x}_0) \neq \mathbf{0}$ then the ***Lagrange multiplier equations*** hold:

$$\nabla f(\mathbf{x}_0) = \lambda \nabla g(\mathbf{x}_0)$$

for a constant λ, the ***multiplier***.

2. The idea of the proof is to use the fact that f has a critical point along any curve in the level set through $\mathbf{x}_0$, which shows, via the chain rule, that $\nabla f(\mathbf{x}_0)$ is perpendicular to that level set; but $\nabla g(\mathbf{x}_0)$ is also perpendicular, so these two vectors are parallel.

3. The Lagrange multiplier method produces *candidates* for extrema; one must make sure there is an extremum and then f can be evaluated at the candidates to choose the maximum or minimum as desired.

4. If there are k constraints

$$g_1 = c_1, \cdots, g_k = c_k,$$

for C^1 functions $g(x_1, \ldots, x_n), \ldots g_k(x_1, \ldots, x_n)$ and constants $c_1, \ldots, c_k$, then the Lagrange multiplier equations become

$$\nabla f(\mathbf{x}_0) = \lambda_1 \nabla g(\mathbf{x}_0) + \cdots + \lambda_k \nabla g(\mathbf{x}_0).$$

5. The Lagrange multiplier method is an effective tool for finding the extrema of $f|\partial U$ in the strategy for finding global extrema described in the last section.

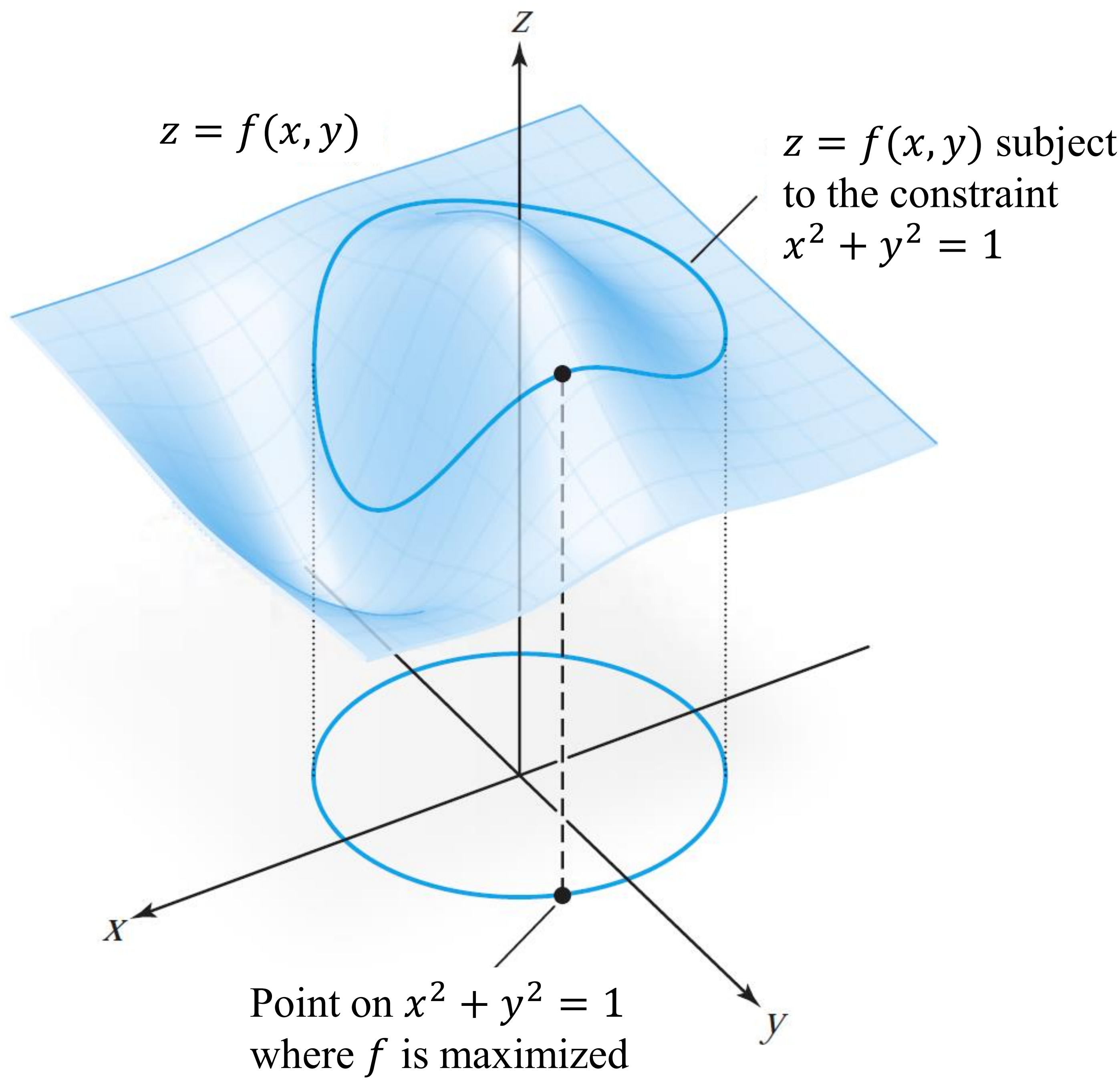
6. **Second Derivative Test with Constraints.** Let $\mathbf{x}_0$ satisfy the conditions of the Lagrange multiplier theorem (in point 1.) Let $h = f - \lambda g$ and $|\bar{H}|$ be the ***bordered Hessian determinant***:

$$|\bar{H}| = \begin{vmatrix} 0 & -\dfrac{\partial g}{\partial x} & -\dfrac{\partial g}{\partial y} \\ -\dfrac{\partial g}{\partial x} & \dfrac{\partial^2 h}{\partial x^2} & \dfrac{\partial^2 h}{\partial x \partial y} \\ -\dfrac{\partial g}{\partial y} & \dfrac{\partial^2 h}{\partial x \partial y} & \dfrac{\partial^2 h}{\partial y^2} \end{vmatrix}$$

evaluated at $\mathbf{x}_0$.

If $|\bar{H}| > 0$, then $\mathbf{x}_0$ is a local maximum of f subject to the constraint $g = c$ and if $|\bar{H}| < 0$, it is a local minimum.

$f\colon U \subset \mathbb{R}^2 \to \mathbb{R}$ and $g\colon U \subset \mathbb{R}^2 \to \mathbb{R}$



z
$z = f(x, y)$
$z = f(x, y)$ subject
to the constraint
$x^2 + y^2 = 1$
x
y
Point on $x^2 + y^2 = 1$
where f is maximized

THEOREM 8: The Method of Lagrange Multipliers Suppose that $f: U \subset \mathbb{R}^n \to \mathbb{R}$ and $g: U \subset \mathbb{R}^n \to \mathbb{R}$ are given C^1 real-valued functions. Let $\mathbf{x}_0 \in U$ and $g(\mathbf{x}_0) = c$, and let S be the level set for g with value c (recall that this is the set of points $\mathbf{x} \in \mathbb{R}^n$ satisfying $g(\mathbf{x}) = c$). Assume $\nabla g(\mathbf{x}_0) \neq \mathbf{0}$.

If $f|S$, which denotes "f restricted to S," has a local maximum or minimum on S at $\mathbf{x}_0$, then there is a real number λ such that

$$\nabla f(\mathbf{x}_0) = \lambda \nabla g(\mathbf{x}_0). \tag{1}$$

We know that the tangent hyperplane (T) to S in $\boldsymbol{x}_0$ is perpendicular to $\nabla g(\boldsymbol{x}_0)$

Given a curve $\boldsymbol{c}(t)$ on S such that $\boldsymbol{c}(0) = \boldsymbol{x}_0$, if $f|S$ has an extreme value, then

$$\left.\frac{df(\boldsymbol{c}(t))}{dt}\right|_{t=0} = 0$$

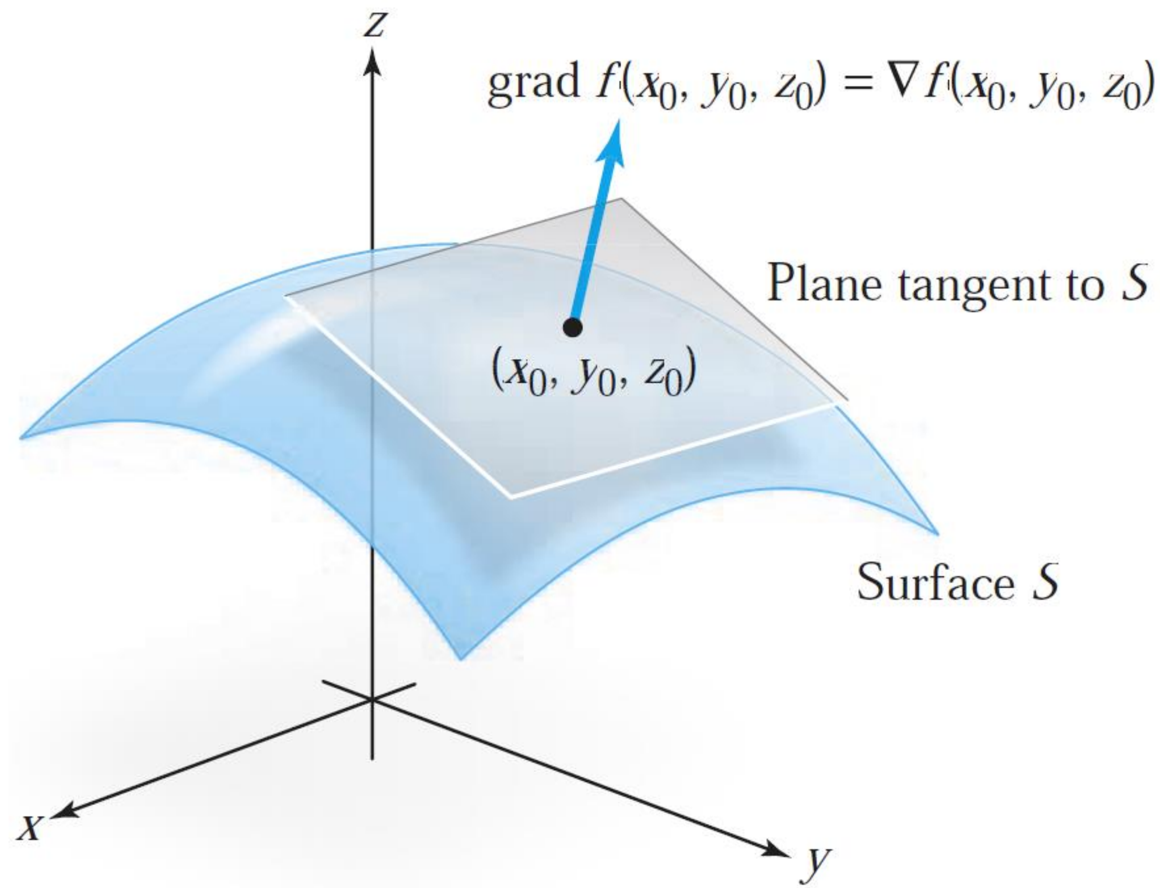
Using the chain rule,

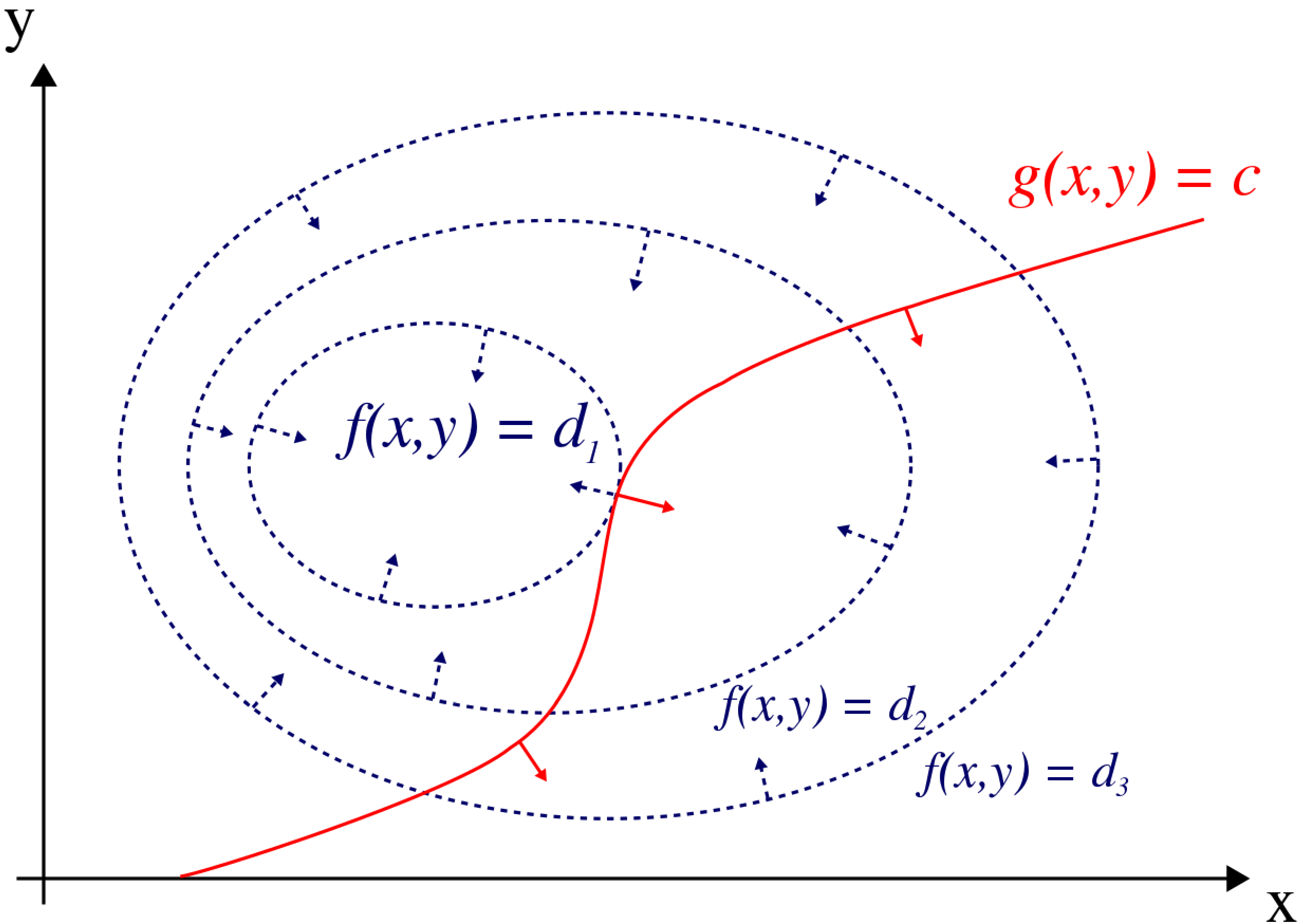
$$\left.\frac{df(\boldsymbol{c}(t))}{dt}\right|_{t=0} = \nabla f(\boldsymbol{x}_0) \cdot \boldsymbol{c}'(0) = 0$$

This means $\nabla f(\boldsymbol{x}_0)$ is perpendicular $\boldsymbol{c}'(0)$, which is a vector in T. Since this is true for all possible curves $\boldsymbol{c}(t)$, then $\nabla f(\boldsymbol{x}_0)$ is perpendicular to T, thus $\nabla f(\boldsymbol{x}_0)$ and $\nabla g(\boldsymbol{x}_0)$ are parallel, and

$$\nabla f(\boldsymbol{x}_0) = \lambda \nabla g(\boldsymbol{x}_0)$$

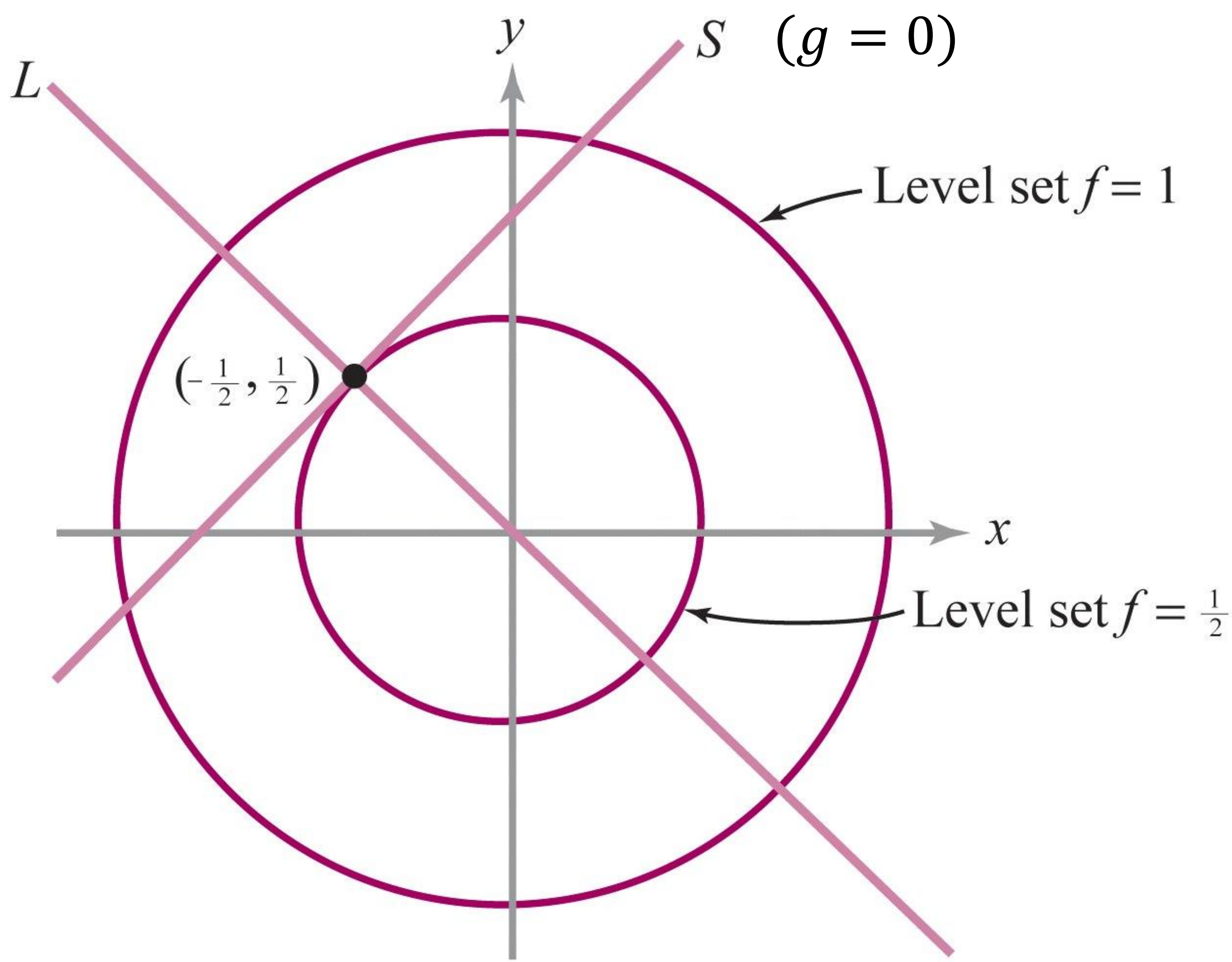
THEOREM 9 If f, when constrained to a surface S, has a maximum or minimum at $\mathbf{x}_0$, then $\nabla f(\mathbf{x}_0)$ is perpendicular to S at $\mathbf{x}_0$ (see Figure 3.4.2).




y
x
$g(x,y) = c$
$f(x,y) = d_1$
$f(x,y) = d_2$
$f(x,y) = d_3$

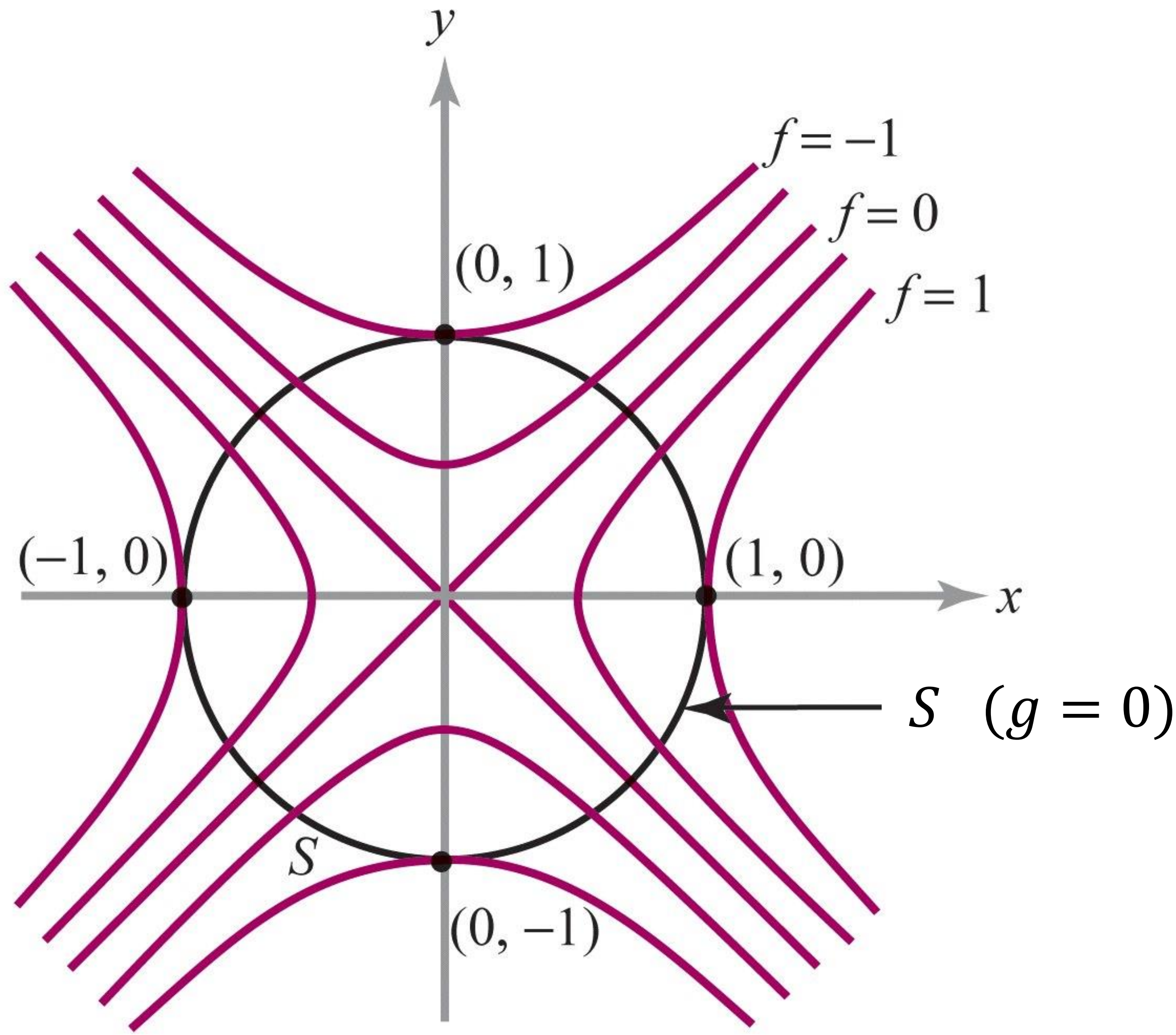
$S = \{(x, y) \in \mathbb{R}^2 \colon y = x + 1\}$ $\quad f(x, y) = x^2 + y^2$

$g(x, y) = y - x - 1$



$S = \{(x, y) \in \mathbb{R}^2 \colon x^2 + y^2 = 1\}$ $\qquad$ $f(x, y) = x^2 - y^2$

$g(x, y) = x^2 + y^2 - 1$



Lagrange function for a single constraint

Function to optimize: $f\colon \mathbb{R}^n \to \mathbb{R}$
Constraint: $g(\boldsymbol{x}) = 0$, where $g\colon \mathbb{R}^n \to \mathbb{R}$

Lagrange function

$$\boxed{L(\boldsymbol{x}, \lambda) = f(\boldsymbol{x}) - \lambda g(\boldsymbol{x})}$$

Find (smooth) critical points by solving the system

$$\frac{\partial L}{\partial x_i} = 0, \qquad i = 1, \dots, n$$
$$\frac{\partial L}{\partial \lambda} = 0$$

Lagrange function for a single constraint

Function to optimize: $f\colon \mathbb{R}^n \to \mathbb{R}$
Constraint: $g(\boldsymbol{x}) = 0$, where $g\colon \mathbb{R}^n \to \mathbb{R}$

Lagrange function

$$\boxed{L(\boldsymbol{x}, \lambda) = f(\boldsymbol{x}) - \lambda g(\boldsymbol{x})}$$

Find (smooth) critical points by solving the system

$$\frac{\partial L}{\partial x_i} = 0, \qquad i = 1, \dots, n \qquad \Leftrightarrow \qquad \nabla f(\boldsymbol{x}) = \lambda \nabla g(\boldsymbol{x})$$

$$\frac{\partial L}{\partial \lambda} = 0 \qquad \Leftrightarrow \qquad g(\boldsymbol{x}) = 0$$

Lagrange function for multiple constraints

Function to optimize: $f: \mathbb{R}^n \to \mathbb{R}$
Constraints: $g_k(\boldsymbol{x}) = 0$, where $g_k: \mathbb{R}^n \to \mathbb{R}$, $k = 1, \dots, C$

Lagrange function

$$\boxed{L(\boldsymbol{x}, \lambda) = f(\boldsymbol{x}) - \sum_{k=1}^{C} \lambda_k \, g_k(\boldsymbol{x})}$$

Find (smooth) critical points by solving the system

$$\frac{\partial L}{\partial x_i} = 0, \qquad i = 1, \dots, n \quad \Leftrightarrow \quad \nabla f(\boldsymbol{x}) = \sum_{k=1}^{C} \lambda_k \nabla g_k(\boldsymbol{x})$$

$$\frac{\partial L}{\partial \lambda_k} = 0, \qquad k = 1, \dots, C \quad \Leftrightarrow \quad g_k(\boldsymbol{x}) = 0$$

Suppose one wants to find the maximum values of

$$f(x,y) = x^2 y$$

with the condition that the x- and y-coordinates lie on the circle around the origin with radius $\sqrt{3}$. That is, subject to the constraint

$$g(x,y) = x^2 + y^2 - 3 = 0.$$

As there is just a single constraint, there is a single multiplier, say λ.

The constraint $g(x,y)$ is identically zero on the circle of radius $\sqrt{3}$. Any multiple of $g(x,y)$ may be added to $g(x,y)$ leaving $g(x,y)$ unchanged in the region of interest (on the circle where our original constraint is satisfied).

Applying the ordinary Lagrange multiplier method yields

$$\begin{aligned}\mathcal{L}(x,y,\lambda) &= f(x,y) + \lambda \cdot g(x,y) \\ &= x^2 y + \lambda(x^2 + y^2 - 3).\end{aligned}$$

from which the gradient can be calculated:

$$\begin{aligned}\nabla_{x,y,\lambda}\mathcal{L}(x,y,\lambda) &= \left(\frac{\partial \mathcal{L}}{\partial x}, \frac{\partial \mathcal{L}}{\partial y}, \frac{\partial \mathcal{L}}{\partial \lambda}\right) \\ &= \left(2xy + 2\lambda x, x^2 + 2\lambda y, x^2 + y^2 - 3\right).\end{aligned}$$

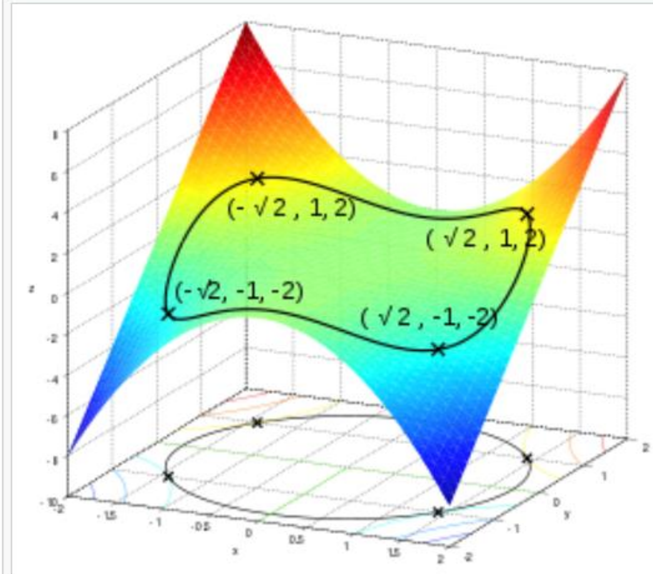


Illustration of the constrained optimization problem

And therefore:

$$\nabla_{x,y,\lambda}\mathcal{L}(x,y,\lambda)=0 \quad \iff \quad \begin{cases} 2xy+2\lambda x=0 \\ x^2+2\lambda y=0 \\ x^2+y^2-3=0 \end{cases} \quad \iff \quad \begin{cases} x(y+\lambda)=0 & \text{(i)} \\ x^2=-2\lambda y & \text{(ii)} \\ x^2+y^2=3 & \text{(iii)} \end{cases}$$

(iii) is just the original constraint. (i) implies $x=0$ *or* $\lambda=-y$. If $x=0$ then $y=\pm\sqrt{3}$ by (iii) and consequently $\lambda=0$ from (ii). If $\lambda=-y$, substituting this into (ii) yields $x^2=2y^2$. Substituting this into (iii) and solving for y gives $y=\pm 1$. Thus there are six critical points of $\mathcal{L}$:

$$(\sqrt{2},1,-1); \quad (-\sqrt{2},1,-1); \quad (\sqrt{2},-1,1); \quad (-\sqrt{2},-1,1); \quad (0,\sqrt{3},0); \quad (0,-\sqrt{3},0).$$

Evaluating the objective at these points, one finds that

$$f(\pm\sqrt{2},1)=2; \quad f(\pm\sqrt{2},-1)=-2; \quad f(0,\pm\sqrt{3})=0.$$

Therefore, the objective function attains the global maximum (subject to the constraints) at $(\pm\sqrt{2},1)$ and the global minimum at $(\pm\sqrt{2},-1)$. The point $(0,\sqrt{3})$ is a local minimum of f and $(0,-\sqrt{3})$ is a local maximum of f.

Troba els extrems de

$$f(x, y, z) = x + y + z$$

amb els lligams

$$x^2 + y^2 = 2$$

$$x + z = 1$$

DEFINITION Let U be an open region in $\mathbb{R}^n$ with boundary ∂U. We say that ∂U is ***smooth*** if ∂U is the level set of a smooth function g whose gradient ∇g never vanishes (i.e., $\nabla g \neq \mathbf{0}$). Then we have the following strategy.

Lagrange Multiplier Strategy for Finding Absolute Maxima and Minima on Regions with Boundary Let f be a differentiable function on a closed and bounded region $D = U \cup \partial U, U$ open in $\mathbb{R}^n$, with smooth boundary ∂U.

To find the absolute maximum and minimum of f on D:

(i) Locate all critical points of f in U.

(ii) Use the method of Lagrange multiplier to locate all the critical points of $f|\partial U$.

(iii) Compare the value of f at all of these critical points.

(iv) Compute all of these values and select the largest and the smallest.

Troba els màxims i mínims absoluts de

$$f(x, y, z) = x^2 + y^2 + z^2 - x + y$$

en la bola D definida per $x^2 + y^2 + z^2 \leq 1$

Troba els màxims i mínims absoluts de

$$f(x,y,z) = x^2 + y^2 + z^2 - x + y$$

en la bola D definida per $x^2 + y^2 + z^2 \leq 1$

$$D = U \cup \partial U$$

$$U = \{(x, y, z) \mid x^2 + y^2 + z^2 < 1\}$$

$$\partial U = \{(x, y, z) \mid x^2 + y^2 + z^2 = 1\}$$

THEOREM 10 Let $f\colon U \subset \mathbb{R}^2 \to \mathbb{R}$ and $g\colon U \subset \mathbb{R}^2 \to \mathbb{R}$ be smooth (at least C^2) functions. Let $\mathbf{v}_0 \in U$, $g(\mathbf{v}_0) = c$, and S be the level curve for g with value c. Assume that $\nabla g(\mathbf{v}_0) \neq \mathbf{0}$ and that there is a real number λ such that $\nabla f(\mathbf{v}_0) = \lambda \nabla g(\mathbf{v}_0)$. Form the auxiliary function $h = f - \lambda g$ and the ***bordered Hessian*** determinant

$$|\bar{H}| = \begin{vmatrix} 0 & -\dfrac{\partial g}{\partial x} & -\dfrac{\partial g}{\partial y} \\ -\dfrac{\partial g}{\partial x} & \dfrac{\partial^2 h}{\partial x^2} & \dfrac{\partial^2 h}{\partial x\,\partial y} \\ -\dfrac{\partial g}{\partial y} & \dfrac{\partial^2 h}{\partial x\,\partial y} & \dfrac{\partial^2 h}{\partial y^2} \end{vmatrix} \quad \text{evaluated at } \mathbf{v}_0.$$

(i) If $|\bar{H}| > 0$, then $\mathbf{v}_0$ is a local maximum point for $f|S$.

(ii) If $|\bar{H}| < 0$, then $\mathbf{v}_0$ is a local minimum point for $f|S$.

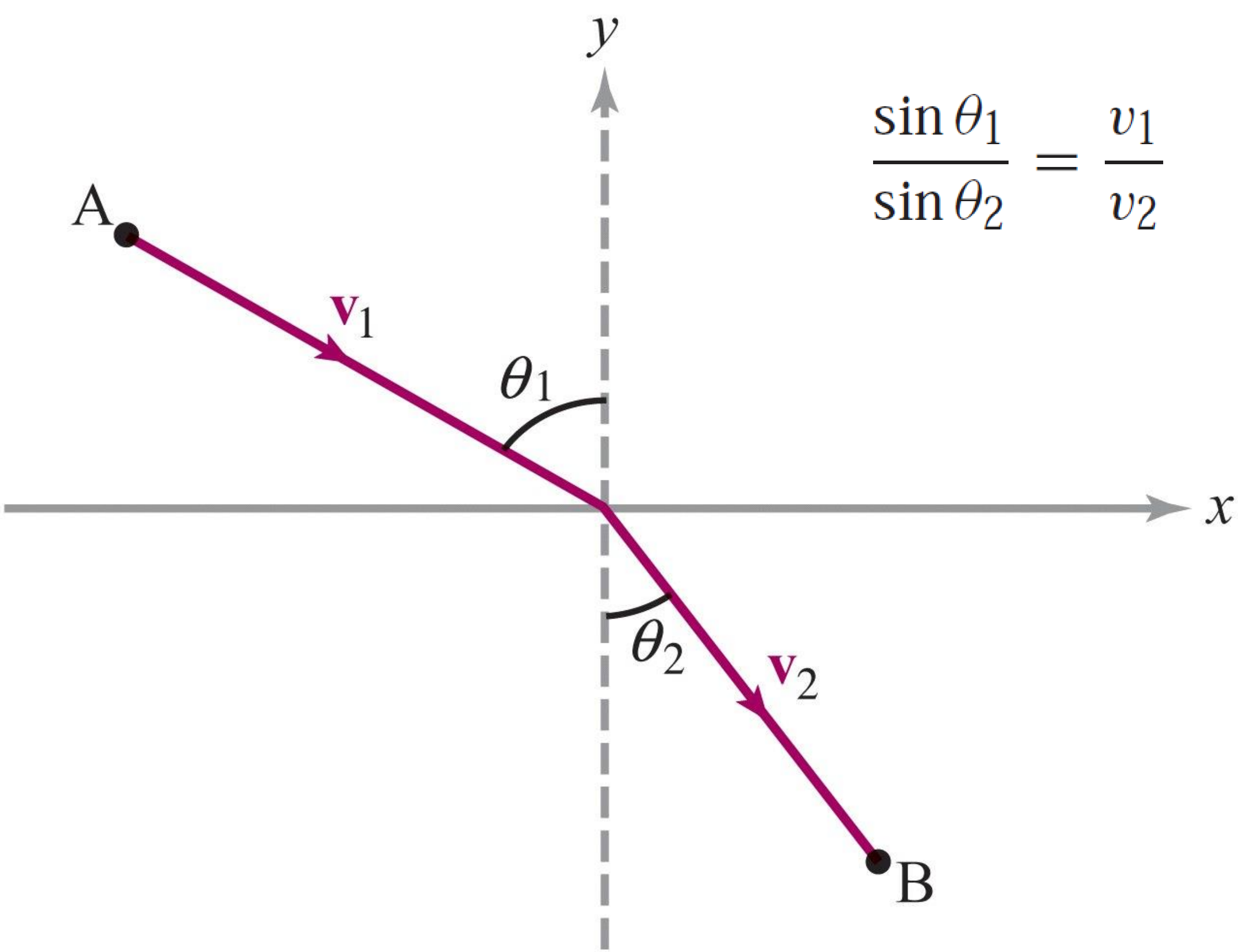
(iii) If $|\bar{H}| = 0$, the test is inconclusive and $\mathbf{v}_0$ may be a minimum, a maximum, or neither.

Troba els extrems de

$$f(x, y) = (x - y)^n, \qquad n \geq 1$$

amb el lligam $x^2 + y^2 = 1$, i classifica'ls utilitzant la matriu hessiana orlada (bordered hessian matrix)

Sabent que la llum viatja a velocitat v_1 si $y > 0$ i a velocitat v_2 si $y < 0$, demostra que el camí que minimitza el temps d'anar entre un punt A ($y > 0$) i un punt B ($y < 0$) satisfà la Llei de Snell



$$\frac{\sin\theta_1}{\sin\theta_2} = \frac{v_1}{v_2}$$